Bisrate Tadesse

AI Engineer & Full Stack Developer — LLM Applications, AI Agents & Machine Learning

Addis Ababa, Ethiopia | +251 967 044 111 | fikertetadesse1403@gmail.com
linkedin.com/in/bisrate-tadesse | bisrat-tadess.com | x.com/BisratTadesse18

PROFILE

Full stack developer with 4+ years of experience building production web and enterprise systems (Python, .NET, JavaScript), now focused on AI engineering. I design and ship LLM-powered applications and autonomous multi-agent systems, and run machine-learning experiments with rigorous walk-forward validation. Particularly interested in adapting AI to low-resource settings and local languages such as Amharic. Combines software engineering depth with an MBA in project management to take AI products from idea to deployment.

CORE SKILLS

AI / LLM engineering	LLM APIs (Gemini, Claude), prompt engineering, structured JSON outputs, multi-agent orchestration, tool use, human-in-the-loop design, retrieval of web content for LLMs
Machine learning	XGBoost, LSTM, meta-labeling, feature engineering, walk-forward validation, backtesting, pandas, NumPy, scikit-learn workflows
Automation	Playwright browser automation, Telegram bots (Bot API, Telethon), scheduled and long-running agents, MetaTrader 5 API
Backend	Python, Django, C# / .NET, Node.js, Express.js, REST API design and integration
Frontend	JavaScript, TypeScript, React.js, responsive UI
Data & DevOps	MySQL, MongoDB, SQLite, Git, cloud services, performance optimisation

AI & MACHINE LEARNING PROJECTS

Autonomous Multi-Agent Application System

Python · Gemini · Playwright

- Designed a team of cooperating AI agents (scout, evaluator, verifier, writer/reviewer, applicant, assistant) coordinated by an orchestrator with shared SQLite memory.
- Agents search the web, verify sources against official pages, draft and self-critique written answers, and complete multi-page web forms with Playwright.
- Human-in-the-loop via a two-way Telegram bot: agents ask for missing facts and resume automatically; built-in safeguards (daily caps, retry limits, never inventing facts, stopping at CAPTCHAs and payments).

Sira AI — AI Recruiting Agent for the Ethiopian Labour Market

LLM agents · Full stack

- Built an AI recruiting agent dashboard adapting an enterprise HR specification to Ethiopian labour law and local job platforms.
- Supports bilingual Amharic/English communication with candidates and employers.

Machine-Learning Trading Research (forex_ai_trader)

XGBoost · LSTM · walk-forward

- Ran 15 walk-forward experiments on 5 years of hourly data across 21 currency pairs, comparing per-pair and pooled XGBoost models, meta-labeling, cross-pair features, multiple timeframes and LSTM.
- Identified an H4 strategy filtered by a pooled XGBoost win-probability model as the only robust edge; ruled out scalping and deep-learning approaches on evidence.

Automated Trading System & Signal Bot

Python · MetaTrader 5 · Telethon

- Built and operate a multi-engine automated trading system on a live MT5 account, plus backtesters, a 29-pair market-direction tool and a live web dashboard with a JSON feed.
- Developed a Telegram signal bot (Telethon) that parses trade signals from a public channel and executes them in MT5.

WORK EXPERIENCE

Full Stack Developer — Ethiocube Systems

Nov 2023 – Present

Addis Ababa, Ethiopia

- Built and optimised the CubeHCM and CubeORM enterprise products, reducing system response time by 30% and improving database efficiency.
- Developed and maintained server-side applications in Python and .NET, improving backend stability by 40%.

- Integrated multiple front-end and back-end systems, improving API response time by 50%.
- Designed user-friendly JavaScript interfaces that increased user engagement by 25%; shipped new features that raised adoption and retention by 15%.
- Led troubleshooting of production issues, reducing downtime by 35%, and worked in agile cross-functional teams that improved delivery speed by 20%.

Full Stack Web Developer — Libanos Furniture PLC *Feb 2022 – Dec 2022*

Addis Ababa, Ethiopia

- Designed, developed and deployed a fully responsive corporate website, increasing customer engagement by 50%.
- Engineered an interactive product catalogue with dynamic filtering, raising user retention by 35%.
- Cut page load times by 40%, driving a 25% increase in mobile traffic; implemented HTTPS, input validation and authentication.

EDUCATION & TRAINING

Software Engineering Programme — ALX Africa *Aug 2023 – Oct 2024*

Addis Ababa, Ethiopia

MBA, Project Management — Sikkim Manipal University *Jan 2009 – Jul 2011*

Addis Ababa, Ethiopia

LANGUAGES

Amharic (native) · English (professional working proficiency)